LSRII (Sacramento) Standard Operating Procedure & User Guide - DIVA Acquisition and System Shutdown
UC Davis Flow Cytometry Shared Resource

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Diva Acquisition SOP – LSRII

1. Do the fluidic preparation and CST run for the giving instrument (See LSRII STARTUP SOP)
2. Delete previous experiment data (make sure you have a backup of data in your lab) from Diva browser and export folder (D:\BD Export\FCS\"your lab name"\"your name") - **You will be charged instrument time if facility staff has to do this for you.**
3. Once in Diva Select 'Use CST settings"
4. Create a new experiment – blank
    a. Name experiment with name and date
5. Create a new specimen and label the specimen
6. Expand the specimen (click the '+' sign) to expose the connected tube
7. Select (not highlight!) the tube (with the current tube pointer)
    a. This will activate the green tab "Parameters" in the Cytometer window
8. Click the parameters tab and select the best label for each of your fluor chromes.
9. Select the "W" and "H" checkboxes for FSC and SSC.
10. Select "log" checkbox for SSC.
11. Delete or remove any unused parameters (fluorochromes) from the parameter list.
    a. Make sure you click the small black/grey dot to the right of the channel to select it and then hold ctrl to select multiple, once all unwanted parameters are selected click the delete button in the cytometer window.
    b. Note: Sometimes bringing extra parameters along can be useful ask Bridget or Jonathan if this makes sense for you
12. Annotate each parameter with marker used.
    a. **Experiment** (menu at top)–> **experiment layout** (towards the bottom)
    b. Select each detector/marker and type the label (Simply select and start typing)
        i. Note: when running samples if you press "next tube" in the acquisition dashboard when the tube pointer is on the last tube in a specimen it will create a new blank tube but copy all labels from the previous tube. This means you can add labels to only one tube and have them transfer down to the next blank tube with the "next tube" button.
13. Generate the global worksheet.



(located in worksheet window toolbar)

1-Bivariate dot plot / 2-Bivariate contour plot / 3-Histogram

a. Create the following Bivariate dot plot (type 1)
    i. FSC-A X SSC-A (morphology)
    ii. SSC-H X SSC-W (Doublet discrimination)
    iii. live/dead plot (yes every experiment should have a live/dead dye or marker to distinguish live cells from dead! – Sorry, FSC-A X SSC-A plot is NOT enough!)
b. Now create bivariate/contour dot plots or histograms for each remaining color in your panel. You need to have EVERY color represented by at least one plot to properly setup the system. In almost all cases you will want to use the "-A" or area parameter for each channel.

c. Right click on any whitespace in a plot and select "Show Population Hierarchy".

The next two steps are run WITHOUT recording any data!

14. Run the Unstained sample and "find your cells" by establishing the correct Forward Scatter - cell size (FSC) and Side Scatter- cell internal complexity (SSC) voltages for your cell type.
    a. FSC guidelines:
        i. The left edge of the smallest populations of interest should have its FSC signal @ or above 10,000 (the first tick box on a linear FSC axis) when the default FSC-A threshold of 5000 is used
        ii. The right edge of the largest populations of interest should have its FSC signal @ or below 262,144 (not off scale)
    b. SSC guidelines:
        i. Log: The center of mass of all populations should be just below 10^4 on the SSC-A axis.
        ii. Linear: Adjust SSC to ensure the edge of lowest SSC population is @ or above 10,000.
15. Run the full stained samples that are brightest for all of your fluorescent markers to ensure that all of your fluorescent markers are on scale and in the linear range (less than 10^5). *Note: This can require multiple full stained sample types to represent the brightest signals for all markers.*
    a. Adjust each fluorescent detector voltage individually until all events/cells are less than 10^5. *Please note that increasing your voltage above the CST recommended level is OK but remember you are generally not really increasing your separation between populations in any useful way but instead are just moving populations for visual benefit.*
16. Run each single color control and make sure each color is brightest in its own channel (e.g FITC is brightest in FITC)..
    a. If a given marker is not bright in its own channel, either increase the voltage of the primary detector or decrease the voltage of the brighter secondary detector. *Note: It is always better to increase the primary detector's voltage instead of decreasing the secondary detector's voltage, but be careful that you do not push your full stained sample's events/cells above 10^5 with this increase!*
    b. FYI: Step 16 does not affect cytometer data quality but does help with data display and helps to take maximum advantage of some automation features in Flowjo and other analysis packages.

Now you will start recording data

17. Establish # of events to record – acquisition dashboard
18. Now starting with your first tube
    a. Label the tube object in Diva.
    b. Prep and Load sample tube onto the system.
    c. Press the “Acquire” button in the software.
    d. Wait 5 seconds to allow the flow to stabilize.
    e. Press the “Record” button.
    f. Once completed, remove the sample tube and replace it with a tube of DI water.
        i. Leave the sample stage to the side
    g. Click ‘Next Tube” in the Acquisition Dashboard
    h. Continue or Finish:
        i. Continue: return to step a.
        ii. Finished: Close the sample stage, put the system in standby and proceed to export data and cleanup.

Export Data

19. When finished recording data, export all data files(as FCS 3.0) to D:\BD Export\FCS\”your lab name”\”your name” – If your lab does not have a folder please create one.
20. Once the export is complete transfer the exported data to a travel/USB drive. *note: We cannot storage or guarantee data safety so please backup your data ASAP.
21. On your next visit DELETE the data from your previous run/s– make sure you have it backed up in your lab!

***note: Any data older than one month will be deleted, if flow core personal have to remove/delete your data there will be an instrument charge for the time require to do so.**

Shutdown Procedure

Overview:

- Fill tubes with cleaning solutions only 2/3 full, avoid wetting o-ring gasket at top of sample injection port
- **All users must run the cleaning routine!**
- **All cleaning tubes should be made fresh and discarded after use!**
- Run system on High: Run 10% bleach to sanitize and remove residual fluorochrome; follow with 10% Contrad; follow with diH2O to flush the system of any residue.

1. **Go the physical LSRII control panel** (front of instrument)
    a. press “High” button to set sample injection rate
    b. Install a tube of 3mL 10% bleach on sample injection port **(make fresh every time!)**
    c. **For 1 minute**: Move the sample arm to the side (vacuum on, cleaning solution going into waste)

    d. **For 5 minutes**: reengage sample support arm (vacuum off, cleaning solution going into flow cell)
    e. Repeat with 10% Contrad, then 100% diH20
2. **Refill the sheath tank to the weld line – Every time you run please!**
3. **Check the waste level**, if it is more than 50% full, empty the waste by dumping into the sink, refilling with 1L 100% bleach and then reinstall on the system.
4. Verify that no other users are signed up after your appointment (within 2hrs)
    a. If Yes:
        i. Place sample fluidics in “Standby” and log out of Diva but leave the system powered on
    b. If No:
        i. Turn off main power on right side of cytometer
        i. Exit software and leave the computer at the Desktop (Do not log out of windows).